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APPLIED RESEARCH

Letter and Person Recognition in Freeform Air-Writing Using Machine Learning Algorithms

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ABSTRACT Air-writing is an emerging form of human-computer interaction that enables text entry through hand movements in the air. This paper explores air-writing-based person recognition alongside letter recognition to analyze the correlation between subjects and their writing styles. This analysis has potential applications in early childhood education for assessing cognitive and psychomotor skills, as well as in user authentication. A comprehensive framework for freeform air-writing analysis using a wearable glove is introduced, encompassing person and letter recognition, letter segmentation, feature extraction, and dataset generation. Unlike user authentication where person recognition depends on predefined content, such as a passcode or signature, person recognition here depends on the underlying writing style of the writer in free-form. The free-form air-writing indicates the user freedom to select any letter with any size (varying from a quarter to the full size of an A4 page) to write on the air. In the study, a dataset is also developed, containing the air-writing signals from the wearable glove, integrated with IMU sensors (gyroscope and accelerometer). Fourier and wavelet transforms are used to extract features and the performances of various machine learning algorithms, namely Decision Tree, Random-Forest, K-Nearest Neighbors, Support Vector Machine, Artificial Neural Networks, and SubSpace KNN, are comparatively studied. To the best of the authors' knowledge, there is no study for person recognition from freeform air-writing letters through IMU sensors, and also there is no study using the Fourier and Wavelet features in this context. Furthermore, the study is also original due to its publicly available air-writing dataset on the Turkish alphabet and also applying various machine learning algorithms. The experimental results show that SubSpace KNN is superior to the others under the suggested parameter settings.

INDEX TERMS Air writing, person recognition, Arduino dataset, Fourier transform, neural networks, subspace KNN.

I. INTRODUCTION

In today's world, data entry into computers is typically facilitated through keyboards or by using a stylus or finger on touch screens. Traditional data entry methods pose numerous practical issues such as constraints related to the writer's location, limitations of the writable area, difficulties of typing on a keyboard, and challenges faced by individuals with disabilities. For example, a bedridden patient would find data entry via a keyboard challenging. Similarly, writing in confined spaces can be difficult for people experiencing sensory loss or motor function impairment. To solve these problems, various methods for air writing have been proposed in recent times [1]. Air writing recognition refers to the automatic detection of letters written in the air by hand. Air writing studies are primarily envisioned for use in fields such as education, art, technology, communication, media, and computer games, with particular emphasis on security and healthcare. This technology offers several advantages, especially in the education of young age groups and in enhancing psychomotor skills by enabling free hand movements, making such trainings more enjoyable, and eliminating the restrictions of the writing area. Additionally, it provides a life-easing technology for people with disabilities. Air writing recognition is

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a technology developing parallel to wearable technology and human-computer interaction.

Many systems based on human-computer interaction are expanding the use of hand gestures, in addition to voice and visual cues, to enhance interaction. Furthermore, systems sensitive to visual signals controlled by head and other body movements are also being developed. Thanks to these advancements, the increasing interest in touch panels, along with progressively smarter phones, has led to a rise in the number of systems operated by hand or finger gestures [2]. Technologies utilizing hand gestures as a data input, through devices and sensors mounted on mobile devices or garments, are being introduced. With such technological advancements, smartphones, tablets, and many other new technologies have become an integral part of our lives. These devices are used with virtual keyboards and invisible pointers known as virtual mice, operated from small screens. We use these devices to record significant aspects of our lives. For instance, in situations where pen and paper are unavailable or when it's impractical to write on a phone – like while walking outdoors or resting – it becomes very practical and convenient to take notes and have them recorded somewhere [3]. For these reasons, research and studies on air writing are increasingly proliferating.

In air writing studies, various sensors are employed to facilitate human-computer interaction. These studies can be broadly categorized into two types: camera-based systems and motion sensor-based systems. Camera-based systems are currently more prevalent due to easier accessibility of hardware and the lack of need for additional wearable technologies. However, hand movements need to be within the camera's frame, restricting the user to a specific area [3]. Another method proposed for air-writing based human-computer interaction is wearable glove technology. In glove-based studies, hand movements are tracked through sensors integrated into the glove or finger, with data transmitted to a computer either wired or wirelessly. These studies include the recognition of letters written in the air [4], word recognition [5], and individual recognition [6] as subcategories. A common feature of these studies is the creation of a dataset from sensor data and classification using machine learning methods. Current literature can be classified according to the type of sensors used to create the dataset. Most studies use 3-axis gyroscope-accelerometer sensors (IMUs) that measure changes in speed over time and operate on the principle of maintaining balance [7], [8]. There are also studies using specially designed pens [9], but even these pens contain gyroscope-accelerometer sensors. Some studies [10]. use IR infrared cameras, like Microsoft Kinect-style devices, which utilize infrared proximity sensors. Developed specifically for this purpose, the Leap Motion device employs infrared cameras and LEDs to effectively and rapidly track hand and finger movements [11], [12]. The LEDs emit an infrared beam, and the cameras use the returning reflections to detect movements occurring in a semi-circular area above them. The major disadvantage of infrared systems is the need for users to specially position and fix these devices. Generally, these systems are preferred in virtual reality games to enhance human-computer interaction. Therefore, air writing datasets in the literature typically use cards termed IMUs, containing 3-axis gyroscope-accelerometer sensors [13], [14]. In air writing studies involving wearable technologies, the dataset is not usually presented as a reference. However, existing systems have aimed to improve success rates by employing different sensor technologies and techniques [15]. Also, there is an increasing interest in applying machine learning techniques to various problems ([16], [17], [18], [19], [20]) including the air-writing recognition. Related work summarizes the literature from the perspective of underlying techniques.

In this paper, person recognition is proposed, in addition to letter recognition, using the proposed air-writing dataset. The dataset is developed as a part of a wearable computing system that allows text entry by writing on the air. For this purpose, MPU6050 sensors (gyroscope and accelerometer) have been integrated into an Arduino board and mounted on a glove to design a wearable device. Through this glove, hand movements are converted into signals, which are segmented into letters and recorded in a dataset labelled with subjects and letters IDs. The dataset contains 10,401 letter samples from 15 subjects. Thus, a single dataset has been used for both person and letter recognition. Signal Features are extracted by Fourier Transform and Wavelet Transform. Then, various machine learning algorithms, namely, Decision Tree, Random Forest, K-Nearest Neighbor (KNN) algorithm, Support Vector Machine, Artificial Neural Networks, and SubSpace KNN (S-KNN) methods are comparatively studied. Cross-validation is used to assess the performance of the algorithms.

The remainder of the paper is structured as follows. Section II provides a review of previous studies, while Section III describes the design of the used glove, the creation of the dataset for characters written in the air, the feature extraction process, and the classification of the data from the conducted tests. Finally, Section IV compares the results of different algorithms and presents recommendations.

II. RELATED WORK

Recent studies on air-writing have grown in diversity and complexity, focusing on different purposes, data generation methods, and evaluation techniques. The first challenge lies in selecting, calibrating, and denoising the sensors that convert writing movements into signals. Devices used include IMU sensors (gyroscopes, accelerometers, magnetometers), vision sensors (cameras, infrared, depth sensors), acoustic sensors (microphones, speakers), radio sensors (WiFi, radar, mmWave), and EMG sensors. EMG signals capture muscle activity during hand movements, classified using machine learning models like CNNs and LSTMs [21]. Radio frequency (RF)-based methods, including WiFi [22], radar [23], and mmWave [24], analyze variations in reflected signal properties caused by hand movements, employing machine

learning or signal processing techniques to recognize patterns corresponding to letters. Similarly, acoustic approaches [25] emit sound waves (often ultrasonic) and analyze their reflections or variations for recognition.

Sensor selection for air-writing depends on the application, user profile (kids, disabled users, etc), and target environment. EMG-based recognition struggles with precise sensor placement, noise sensitivity (e.g., motion artifacts, sweat), and individual muscle variations, requiring personalized calibration. While vision, acoustic, and RF-based approaches offer contactless solutions, they face issues like cost, interference (WiFi), and portability (radar). Acoustic methods are limited by environmental noise and short range. IMU and vision sensors are widely used but still pose challenges like segmenting signals, feature extraction, and character recognition, which remain key focuses in the literature.

Feng and his colleagues, in their study [26], developed a real-time character recognition system for finger writing using Kinect sensor's depth information. They recorded a total of 8000 entries from two different distances for two different people, including numbers, lines, English and Chinese characters, using fingertip tracking. They used the K-means algorithm to divide the segmented image into two parts: one for the finger-hand and the other for the hand-arm. They reported an accuracy of 90% for English characters and around 80% for Chinese characters in real-time tests [26].

Zhang and his colleagues, in their study [27], used a Kinect sensor for another air writing recognition study. They employed a three-layered artificial neural network model with three inputs, 30 neurons in the hidden layer, and nine outputs. The algorithm was trained with Resilient Back Propagation with the sigmoid activation function. They created a dataset of 375 videos, totaling 44,522 frames, written in the air with the fingertips by a five-person team. They reported the ability to recognize the 6,763 most frequently used Chinese characters, all English characters in both uppercase and lowercase, and all digits with over 90% accuracy [27].

Mohammadi and Maleki in their study [28] attempted to recognize Persian characters using the Kinect device. They used and proposed a method called slope variations detection, resistant to rotation changes in the trajectory, to extract features from the trajectory. Additionally, a new analytical classifier was proposed to match a vector to a letter. This classifier was said to be faster and more accurate than Support Vector Machines, Hidden Markov Model, and K-Nearest Neighbor algorithms. The tests showed a 98% recognition rate for Persian numerals.

Kumar and his colleagues, in their study [29], created a database of 320 Latin sentences using the Leap Motion device by air writing with fingertips, divided the sentences into lines based on large spaces at the beginning and end, and then segmented them into words for word recognition using a Hidden Markov Model-based classifier. They achieved a success rate of 80.3% in segmenting into words and an accuracy of 92.73% in word recognition tests. They stated that the results indicate Leap Motion could be a cost-effective data input alternative compared to traditional systems [29].

In another study [30], using the Leap Motion device, Kumar and colleagues introduced 28 words in sign language to a group of 10 users and also had them spell these words with fingers, creating a dataset of 2240 words. They used Support Vector Machines for classifying whether the writing was done by hand or finger, and a Bidirectional Long Short-Term Memory Neural Network (BLSTM-NN) for detecting the writing. In real-time recognition of both types of movements, they achieved an accuracy of 63.57% [30].

Khan and his colleagues, in their study [11], utilized the Leap Motion device to recognize both English letters, numbers and some words written in the air. They generated a dataset, on which they used artificial neural networks to recognize the characters. They report achieving an accuracy rate of over 90% in their study [11].

Bastas and his colleagues, in their study [31], acquired data from 10 individuals who are asked to write each digit, from 0-9, 10 times using the Leap Motion device, thereby generating a dataset of approximately 1200 entries. They employed deep learning architectures like Convolutional Neural Networks (CNN) for classifying these digits and reported an average success rate of 90.20% using CNN [31].

Alam et al. [32] utilized a depth camera to capture fingertip trajectories in three-dimensional space for tracking purposes. They applied nearest neighbor and root point translation methods for feature selection and trajectory normalization. Subsequently, they employed a combination of long short-term memory (LSTM) and convolutional neural networks (CNN) for gesture recognition, achieving an accuracy of 99.32% on the 6D Motion Gesture (6DMG) alphanumeric dataset.

Kumar and his colleagues, in their study [33], used generic webcam to detect and recognize the air-written characters, which reduces the dependence on external hardware. The system does 2D trajectory approximation using HSV color space and CNN to recognize the air-written letters. The CNN is trained on four different datasets and the accuracy scores are for each dataset is reported to be above 99%.

Most previous approaches to air-writing relied on external depth sensors, such as Leap Motion, Kinect. While these methods offer higher recognition accuracy, they come with increased production costs, which restricts their accessibility for general consumers. Thus, IMU-based approaches still preserve their importance with the advantage of cost-effectiveness and easy deployment, such as using a wearable device. Cho and his colleagues, in their study ([34]), collected data through IMU sensors of the mobile phone and used the Bayesian Network for modeling and Support Vector Machines algorithm for classification. They created a dataset of 11 characters, including digits (0-9) and letters (O, X), and aimed to classify them. They achieved a classification accuracy of 96.9% with the Bayesian network. In 2009, Agarwal et al. [35] noted that the keyboard usage on phones was

not well-received by the general population. They proposed PhonePoint pens to use the phone as a pencil to write text on the air in real-time. However, the approach would work with a specific database limited to a fixed matching set.

Amma and his colleagues, in their study [5], designed a special glove, integrated with IMU sensors, and conducted text recognition both dependent and independent of the individual using the Hidden Markov Model. In their study with 10 people, they achieved a character recognition accuracy of 94.8% in individual-dependent recognition and 81.9% in individual-independent recognition. For text recognition, they attained a 97.5% success rate with a vocabulary of 652 words.

Dash and his colleagues [36] introduced AirScript, an air-writing recognition system that combines deep learning with Myo-armband hardware, which includes gyroscope and accelerometer sensors. They employed CNN, two GRU networks, and their fusion to recognize characters written in the air. To benchmark their approach, the digit dataset collected from 12 individuals was also tested using traditional machine learning algorithms like KNN, Support Vector Machines, and Hidden Markov Models. The proposed AirScript achieved accuracy rates of 91.7% for independent recognition and 96.7% for dependent recognition of digit characters [36].

Chen et al. [37] focused on recognizing air-writing using a 6-DOF motion tracking system, which captures position, orientation, acceleration, and angular speed. From the motion data, five basic features were extracted and combined into pure optical, pure inertial, and complete 6-DOF features. In the study, word recognition was modeled using HMM (Hidden Markov Model) for both word-based recognition and letter-based recognition. In latter, motion words were modeled by combining character and ligature models. The study used a custom dataset where each isolated motion character (A to Z) was recorded ten times by 22 subjects and some selected words were prepared similarly. Word-based recognition achieved low word error rates but could not recognize out-of-vocabulary words, making it suitable for applications with limited vocabularies requiring high accuracy. In contrast, letter-based recognition was able to handle arbitrary letter sequences with around 10% accuracy.

Younas et al. [38] introduced a finger-worn IMU-based system for segmenting air-writing activity. The study explored various deep neural network models, including OSCNN (the Omni-scale 1D-CNN), LSTM, Bi-LSTM, ResNet, and GTN (Gate Transformer Network), to analyze spatio-temporal information for air-writing gesture segmentation. A comprehensive dataset was created by collecting air-gesture data from 50 participants. All deep learning models achieved exceptional results using raw sensor data, with the Bi-LSTM model delivering the highest segmentation (binary classification) accuracy of 98.23%.

Younas et al. [39] developed FairWrite, a system that employs finger-worn IMU sensors to recognize air-writing characters. A freeform dataset was collected from 30 participants using this system. FairWrite reconstructs character trajectories by mapping motion data to a 2D canvas using Euler angles. For recognition, KNN combined with DTW (Dynamic Time Warping), 1D-CNNs, and BLSTM were applied to identify letters and digits. The best performance was achieved using KNN with DTW, resulting in an overall accuracy of 94%, with digits recognized at 85.4% and letters at 95.5%.

Luo et al. [40] proposed an IMU-based system for recognizing motion letters using Dynamic Time Warping as the primary recognition algorithm. Data were collected from three users under different scenarios, including single-user and multi-user cases, where participants repeatedly wrote the English alphabet and digits 12 times. In experiments, the system achieved an accuracy of 84.6% for uppercase letters ('A' to 'Z') in the user-dependent (single-user) case. The study also highlighted that the DTW-based method is highly user-dependent.

Tripathi et al. [41] addressed air-writing recognition by converting IMU signals into images, which are then processed using deep learning models. Signals from a 3-axis accelerometer and gyroscope were encoded as images using techniques like self-similarity matrix (SSM), Gramian angular field (GAF), and Markov transition field (MTF) to create two sets of 3-channel images. Several image classification architectures, including ResNet, DenseNet, VGGNet, AlexNet, and GoogleNet, were used. Experiments on two public datasets showed the proposed method's effectiveness, with the best accuracy of 88.43% achieved using a CNN with GAF+DenseNet161 combination.

Tripathi et al. [42] considered an IMU based system for user and device independent air-write recognition. They proposed two stage recognition where the first stage involves training an encoder network with supervised contrastive loss and the second stage involves training the classification head only (the encoder weights kept frozen). 1DCNN, LSTM, BiLSTM, and DCNN-LSTM/BiLSTM were tested for encoder network on two datasets. Both datasets contains motion data from many users through different IMU hardwares. The 1DCNN-BiLSTM model was reported to perform the best among all the model architectures with accuracy of 88.05%.

Ecer et al. [4] used Random Forest, KNN, Decision Tree and Mahalanobis classification to recognize air-written Turkish letters. They created a custom dataset using a wearable glove integrated with an Arduino board and IMU sensors. However, the dataset was limited to data from a single user. They used Fourier coefficients to represent the motion signals. The system achieved a letter identification accuracy of 91.10% on the test set with Random Forest method.

Zhang et al. [43] introduced a real-time wearable system for air-writing recognition using the Arduino Nano 33 BLE Sense as an edge device running TensorFlow Lite. The system allows users to write 36 characters (10 digits and 26 lowercase letters) in free space, leveraging motion data from embedded IMUs processed by a microcontroller. With 63,000

air-writing samples collected from 35 subjects (18 males, 17 females), the system's CNN achieved a recognition accuracy of 97.95%.

Predominantly, the majority of these studies focus on character (letter) recognition, with a smaller portion dedicated to text recognition in limited settings. Also the studies related to person identification are only considered in the context of user authentication [44], [45]. User authentication has its own limitations and assumptions, such as writing a secret text (passcode) in the air a few times during learning stage and require the memorization of predefined special gesture patterns from both content and style perspective. Wijewickrama and his colleagues ([46]) reviewed these studies and concluded that even in the context of user authenticity, these studies have not yet been successfully implemented in real-life despite the growth of wearable technologies like smart wristbands, watches and so on. They stated that handwriting-based identity verification could see significant development in the future with more sensitive and efficient sensors [46]. However, recently air-writing is also proposed to improve the cognitive and psychomotor skills of childs in early education [3], [47], [48]. Based on these studies, it is suggested that person recognition through freeform air-writing could be used as a way to assess how the writing style is personalized over the course of time, which is an act of cognitive development as well. Thus, this study is different from the literature in a few perspectives. First, it introduces person recognition from freeform air-writing letters through IMU motion sensors (specifically focusing on individual writing style recognition rather than user authentication). Secondly, it introduces the usage of Fourier and Wavelet features on the basis of various machine learning algorithms, which, to the best of the authors' knowledge, has not been explored in prior air-writing research. Thirdly, a novel freeform air-writing dataset is developed and made publicly available.

III. MATERIAL AND METHOD

The proposed system model is shown in Fig.1 where each phase is explained in separate subsections subsequently. Given the focus on the limited target platform and interpretability of the individual writing styles, classical machine learning algorithms and a simpler ANN architecture were opted for, rather than deep learning. It is believed that these methods provide a good balance between performance and computational cost where acceptable performance above 91.5% accuracy is achieved.

A. GLOVE DESIGN AND SEGMENTATION OF LETTER SIGNALS

In the study, the MPU 6050 IMU card, which is compatible with Arduino and features a 3-axis accelerometer and a 3-axis gyroscope, was used. This card possesses a 6-axis degree of freedom (DoF).

The glove was developed in a previous study [4] where the Arduino card is positioned on the top of the glove, and the MPU6050 sensor is placed on the fingertip, emulating the act of real writing. The glove is depicted in Fig.2. The device, designed for the air-writing task, is suitable in the hand due to its lightweight and compact size. Also, due to the nature of the air-write experiment, the glove is only used for data acquisition. The analysis of air-write data is considered a post-processing application. This is the reason that simple hardware (Arduino) is considered for data collection from special groups (kids, disabled people, etc.).

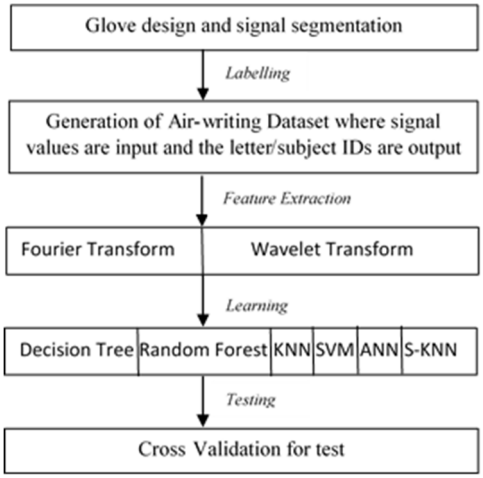


FIGURE 1. System model.

The subjects were instructed to write the letters in start-stop manner to make the extraction of letter signals easier. That is, the subject waits few seconds after writing each letter on the air. Each subject writes the letters several times in a start-stop manner and the device captures the 6-channel data, corresponding to the 6 motion sensors. Each task involves writing a sequence of letters on the air. Figure 3 shows some cut from the segmented signals where each channel data is shown in a different color and the letter segments are contained in rectangles. This figure just shows how the segmented signals look like for a person writing the letter 'A' six times. It is apparent that the signal exhibits high variations during the activity compared to the periods of no activity. This information can be leveraged for segmentation [4]. In the following, a brief description of the letter segmentation algorithm is provided.

Let $\mathbf{c}[n] = [c_1[n], c_2[n], \ldots, c_6[n]]^T \in \mathbb{R}^{6\times 1}$ represent 6-dimensional (6-channel) signal measured at the sampling time t_n from six motion sensors during an arbitrary air-writing task, where $c_i[n]$ denotes the measurement from the i-th motion sensor. The signal is acquired over a fixed period and consists of a total of N samples.

The signal from the i-th channel is first normalized to have zero mean and unit variance, i.e.,

$$c_i[n] \leftarrow \frac{c_i[n] - \mu(c_i)}{\sigma(c_i)}$$

where $\mu(c_i)$ and $\sigma(c_i)$ represent the mean and standard deviation of the i-th channel signal, respectively. The normalized signal is then used to compute another signal, $v_i[n]$, which

TABLE 1. Letters of turkish alphabet.

A	B	C	Ç	D	E	F	G	Ğ	H
I	İ	J	K	L	M	N	O	Ö	P
R	S	Ş	T	U	Ü	V	Y	Z	

measures the weighted variation at time t_n in the i-the channel based on weighted local statistics, i.e.,

$$v_i[n] = \sqrt{\sum_{l=-L}^{L} (c_i[n+l] - m_i[n])^2 w[l]},$$

where $L \in \mathbb{Z}^+$ is a parameter used for defining a local window, $w[l] = (L - |l| + 1)/(L+1)^2$ is the weight parameter, and $m_i[n]$ is the weighted local mean, i.e.,

$$m_i[n] = \sum_{l=-L}^{L} c_i[n+l]\, w[l].$$

The variation signal $v[n]$ at time t_n is then computed by simply averaging across all channels, i.e.,

$$v[n] = \sum_{i=1}^{6} v_i[n]$$

which further normalized as

$$v[n] \leftarrow \frac{v[n] - \min(v)}{\max(v) - \min(v)},$$

where $\min(v)$ and $\max(v)$ represent the minimum and maximum values of v, respectively.

The variation signal is high (close to one) in the activity regions and low (close to zero) in the no-activity regions. Otsu thresholding [49] was applied to the variation signal, v, in order to adaptively label segments that correspond to high-activity regions. Since there is no activity between the air-writing of two characters, a simple thresholding operation effectively segments the characters. This is achieved by first applying median filtering to the binary signal, which represents activity and no-activity regions resulting from the Otsu thresholding, over a window of size L. The start and end of each segment are then identified when the filtered binary signal transitions from zero to one and from one to zero, respectively.

B. DATASET

The dataset comprises 10,401 letter samples, collected from 15 subjects (8 male, 7 female). The letters cover the 29 letters of Turkish alphabet (see Table 1) in capital form. Letter sizes vary from approximately 1/4 to full A4 size. This was to ensure variety and diversity. Such design allows researchers to study air-writing recognition independent of the letter size.

The imbalanced nature of the dataset, where the number of repetitions per letter varies, is intentional to reflect real-world writing scenarios where some letters occur more frequently

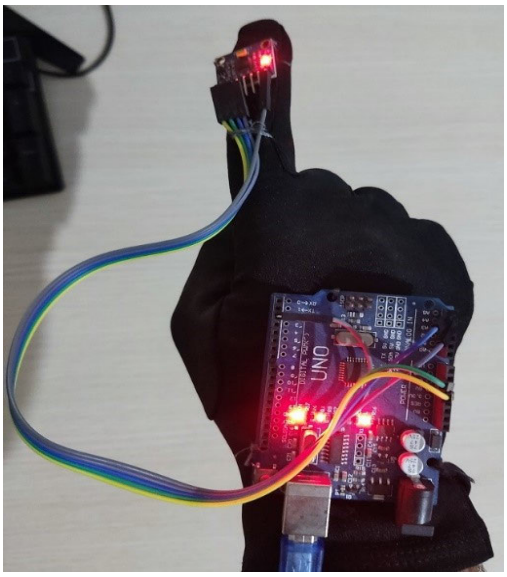


FIGURE 2. The designed glove.

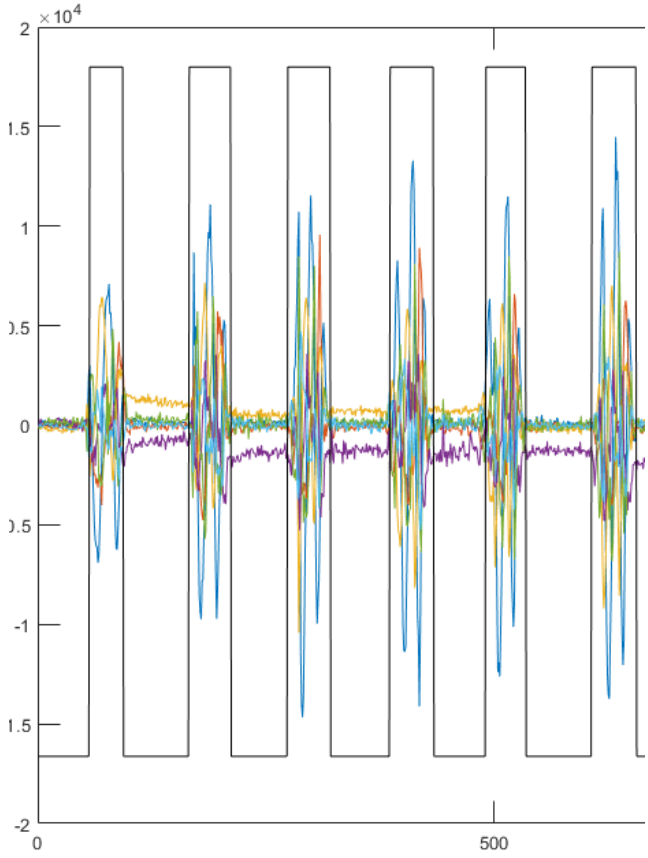


FIGURE 3. A snapshot from the segmentation of 6-channel data.

than others. The MPU6050 sensor used has 16-bit resolution and 100Hz sample rate. The dataset is publicly available at GitHub.[1] The only standard followed is that when writing of

[1] https://github.com/zyetgin/Air-Writing-Dataset

a letter is completed the finger stay fixed for few seconds. This is used as the guard time between the letters to help the segmentation of the letters efficiently from 6-channel data.

Six-channel data from motion sensors are continuously transmitted to the MATLAB™ interface via a USB connection. The segmentation program processes this data to extract letter signals in row format. The dataset will be published in MATLAB™ format as a file named ''dataset.mat''.

This file contains three components:

data: An array holding the segmented motion data.

letter: A target vector containing the letter labels.

user: A target vector containing the user labels.

For each letter k, the corresponding motion sensor data (6)-channel data) is represented as $data_k$ using 6-by-n matrix, where 6 represents the number of sensor channels and n is the number of sample points recorded while writing the letter, which varies from letter to letter.

Thus, the overall representation of the input *data* is as follows;

$$data = \begin{bmatrix} data_1 \\ data_2 \\ data_3 \\ - - - \\ - - - \\ data_k \end{bmatrix} \tag{1}$$

where

$$data_k = \begin{bmatrix} ax_1 ax_2 \ldots\ldots\ldots ax_n \\ ay_1 ay_2 \ldots\ldots\ldots ay_n \\ az_1 az_2 \ldots\ldots\ldots az_n \\ gx_1 gx_2 \ldots\ldots\ldots gx_n \\ gy_1 gy_2 \ldots\ldots\ldots gy_n \\ gz_1 gz_2 \ldots\ldots\ldots gz_n \end{bmatrix}_{6xn} \tag{2}$$

where variables ax, ay, az denotes the sensor values from the accelerometer, gx, gy, and gz are sensor values from the gyroscope, and n is the number of sample points for the particular letter, labelled by $letter_k$.

C. FEATURE EXTRACTION

All 6 channels were used to capture a complete representation of hand motion during air-writing. Two feature extraction methods are employed: the Discrete Fourier Transform (DFT) and the Discrete Wavelet Transform (DWT). The Fourier Transform decomposes data into frequency components, capturing spectral features while disregarding data orientation. This property is particularly valuable in air-writing applications, where human subjects may vary the orientation of their movements across repeated experiments. On the other hand, the Wavelet Transform decomposes data into approximation and detail components. It divides the data into frequency bands, retaining some temporal information in each band. The feature extraction methods are formulated at Eq.4-9 for Fourier Transform and at Eq. 10-14 for Wavelet Transform.

Given the motion sensor data $data_k$ for a particular letter, which is represented as a 6-by-n matrix, the Fourier-based feature extraction method uses the 2-dimensional Fast Fourier Transform (FFT) to convert the input into a fixed-size spectral domain representation. This process is performed via the fft2 function. To ensure the input size is compatible with the fft2 function, the dimensions $6\times$ n are adjusted to 8×128, in powers of two. This adjustment is achieved through zero-padding or truncation at Eq.3, which is implicitly handled by the fft2 function.

$$ResizedMatrix_k = PadOrTruncate(data_k, size = (8, 128)) \tag{3}$$

where PadOrTruncate function ensures efficient computation by resizing the input dimensions to the nearest powers of two.

Then 2D FFT is applied to the resized data to transform it into the spectral domain at Eq.4.

$$fdata_k = fft2(ResizedMatrix_k) \tag{4}$$

where $fdata_k$ represents the fixed-size 8×128 matrix of Fourier coefficients in the spectral domain and fft2 function definition is specified through Eq.5-7.

$$fdata_k[\mathrm{p}+1, \mathrm{q}+1] = \sum_{j=0}^{m-1}\sum_{z=0}^{n-1} \omega_m^{j.p}.\omega_n^{z.q} data_k[\mathrm{j}+1, \mathrm{z}+1] \tag{5}$$

$$\omega_m = e^{-2\pi i/m} \tag{6}$$

$$\omega_n = e^{-2\pi i/n} \tag{7}$$

where i is the imaginary unit, p and j are indices that run from 0 to m–1, and q and z are indices that run from 0 to n–1.

Then the magnitude of the spectral coefficients is computed to obtain the absolute spectral data at Eq.8.

$$|fdata_k| = \sqrt{\Re(fdata_k)^2 + \Im(fdata_k)^2} \tag{8}$$

where $\Re(fdata_k)$ and $\Im\left(fdata_k\right)$ denote the real and imaginary components, respectively.

To generate the feature vector X_k for the corresponding letter $letter_k$, the coefficients of $|fdata_k|$ are serialized in a column-wise manner at Eq.9.

$$X_k = Serialize(|fdata_k|) \tag{9}$$

where Serialize function flattens 8×128 matrix into a single vector of size 1024.

For Wavelet based feature extraction method, first the motion sensor data corresponding to a particular letter, $data_k$, is resized to 6-by-60 matrix using 'nearest neighbor' interpolation method at Eq.10.

$$ResizedMatrix_k = Interpolate(data_k, method =' nearestneighbour') \tag{10}$$

Then, the resized matrix undergoes a 4-level wavelet decomposition using the Daubechies wavelet db4 at Eq.11

where DS is the returning decomposition structure (DS) containing coefficients for horizontal (H), vertical (V), diagonal (D), and approximation (A) components.

$$DS = WaveletDecompose(ResizedMatrix_k, wavelet = db4, level = 4) \quad (11)$$

Then, At the ith decomposition level, the detail coefficients — horizontal (H_i), vertical (V_i) and diagonal (D_i)—are extracted from the decomposition structure at Eq.12.

$$[H_i, V_i, D_i] = ExtractDetails(DS, level = i) \quad (12)$$

The approximation coefficients (A_i) are extracted separately using the Haar wavelet at Eq.13.

$$A_i = ExtractApproximation(DS, wavelet =' haar', level = i) \quad (13)$$

Each coefficient A_i, H_i, V_i, and D_i is a matrix of the same size at i^{th} decomposition level. Finally, all the coefficients are serialized in the order of A, H, V and D to form the feature vector X_k at Eq.14, as shown at the bottom of the page, where first all the coefficients of A_1 matrix are serialized in column-wise manner, then the serialization for other decomposition levels of A follows in order, and then this pattern repeats for the other components in their specified order, where Serialize function is as defined at Eq.9 and the set operator { } concatenates the inputs.

At each level of wavelet decomposition, the column dimension of the matrix is halved. The rows are not halved due to number of rows (6) is less than the db4 filter (8). This halving continues until the specified decomposition level is reached. Each component size is reduced across the levels according to the formulation of the new size = (size + filter_size-1)/2, which produces following sizes: $6 \times 33, 6 \times 20, 6 \times 13, 6 \times 10$. Thus, 456 coefficients are acquired for each component, and $4 \times 456 = 1824$ coefficients are combined to form a single feature vector at the end of final decomposition level.

In summary, the input datasets for the machine learning algorithms, including ANN, are 10401-by-1024 matrix for Fourier based methods and 10401-by-1824 matrix for Wavelet based methods where 10401 is the total number of letter samples.

D. APPLICATION OF MACHINE LEARNING METHODS

1) APPLICATION OF DECISION TREE LEARNING

The CART (Classification and Regression Trees) algorithm is used for Decision Tree Learning. It is used for both classification and regression problems. In the study, CART is used as a classifier to classify letters or subjects. It works by recursively applying binary splitting on the feature space to partition the feature space into smaller regions. Every split creates a node in the tree and minimizes the weighted average of the Gini Index of the left and right partitions. The proposed implementation grows a big tree with the following parameter setting: MinLeafSize = 1, MaxNumSplits = N - 1, and MinParentSize = 10, where N is the number of samples in dataset, MinParentSize is the minimum number of samples in a node for it to become parent, and MinLeafSize is the minimum number of samples for a node to become a leaf, which shapes the final tree.

2) APPLICATION OF RANDOM FOREST LEARNING

Random Forest is an ensemble learning method that combine multiple weak decision trees and take decision by majority voting or averaging for classification and regression respectively. In the study, Random Forest Classification is used to classify letters or subjects. Each decision tree is weakened by training it on a bootstrap dataset, which is generated randomly by sampling with replacement from the original training dataset. Each decision tree is further weakened at node splits during its growing by considering a random subset of features. This randomness helps in creating diverse weak classifiers that collectively form a strong ensemble, reducing the effects of overfitting when a single tree is used and, thus, improve the generalization of the ensemble. The proposed implementation uses 600 decision trees, each deeply grown with the same parameters of the decision tree above. The implementation considers the square root of the number of features as the randomly selected features from the original set at each node split.

3) APPLICATION OF K-NEAREST NEIGHBOR LEARNING

The K-Nearest Neighbor (KNN) algorithm is based on finding the K nearest samples around a particular sample in the feature space where K is the main parameter of the algorithm. Finding the nearest points is another sub problem, which could be achieved by some algorithms, such as KD Tree or Ball Tree. Then, the K nearest neighbors either do majority voting or simple averaging for classification and regression respectively. In the study, KNN classification is used to classify letters or subjects. The KNN assumption is that nearest data points around a sample tend to belong to the same class with the sample. During the training phase of the KNN classification, the approximate positions of the K nearest points around each point in the feature space are coded in a computational structure, e.g. a tree, such that computation of K nearest neighbors around any point in the feature space is possible. The implementation uses. Euclidean distance metric to compute the distance between two points. The distance weight was taken equally. It uses KD Tree to search for the K nearest points with the K value set to 1.

$$X_k = \begin{bmatrix} \{\text{Serialize}\,(A_i)\}_{i=1}^{4}, \{\text{Serialize}\,(H_i)\}_{i=1}^{4}, \{\text{Serialize}\,(V_i)\}_{i=1}^{4}, \\ \{\text{Serialize}\,(D_i)\}_{i=1}^{4} \end{bmatrix} \quad (14)$$

4) APPLICATION OF SUPPORT VECTOR MACHINE LEARNIN

The Support Vector Machine (SVM) algorithm aims to find the optimal hyperplane that maximizes the margin between two classes for classification, or fits a tube with the maximum margin for regression. The core objective is to identify the direction that maximizes this margin, subject to certain constraints, which can be formulated as a constrained optimization problem. In this study, SVM Classification is employed to classify letters or individuals. The standard SVM Classification method uses two margins (+1 and −1) to separate two classes through a linear model, $f(x) = w \cdot x$. However, this binary classification can be extended to multi-class problems using approaches like one-versus-rest (OVR) and one-versus-one (OVO). Additionally, the basic SVM classification can be adapted to tolerate misclassifications (losses), with the degree of tolerance controlled by the parameter, which penalizes the total loss. This parameter balances the trade-off between minimizing the total loss and maximizing the margin, relaxing the requirement for linear separability. Furthermore, SVM models with linear decision boundaries can be applied to non-linear problems by using different kernels, such as Sigmoid, Gaussian, and Polynomial. In the OVR approach, each classifier f_i distinguishes between class i (+1) and all other classes (−1). Thus, a total of models are required, where c is the number of classes. Then, the final decision about the sample x is given by the formula $\text{argmin}_{i=1..c} f_i(x)$.

In the OVO, each model f_{ij} separates between the class i (+1) and class j (−1). Thus, totally c.(c-1)/2 number of models are needed. Then, the final decision about the sample x is given by the majority voting of all models. The proposed implementation uses OVO approach. The parameter is set to 1. The dataset is standard normalized (centered) before training. The cubic kernel is used with box constraint level 1.

5) APPLICATION OF ARTIFICIAL NEURAL NETWORK LEARNING

Artificial Neural Network (ANN) can be used for both classification and regression problems. Originally, it is the Multi-Layer Perceptron where there are many perceptron or neurons organized into communicating layers, which altogether constructs the architecture of the ANN model. The neurons have links to other neurons where the links have weights. The architecture of a particular neural network mainly includes the number of layers, number of neurons in each layer, activation functions, shaping the final output of the layers as well as other settings such as initialization of the weights. Similar to SVM, ANN algorithms search for the best weights that minimizes the total losses. Thus, various ANN models can use different optimization algorithms as backend functionality to optimize the weights. Also, various ANN architectures have their own limitations, advantages and disadvantages, each suited to different types of problems and data. The ANNs are so popular in literature that they are the fundamentals for deep learning. Some popular ANN architectures include Feedforward Neural Networks (FNNs), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) Networks, Autoencoder Networks, Generative Adversarial Networks (GANs), and Transformers.

FNNs are the simplest form of ANNs, also known as Multi-Layer Perceptron, which is suitable for function approximation. CNNs consist of convolutional layers for feature extraction, pooling layers for down-sampling, and fully connected layers for classification/regression. CNNs are suitable for processing grid-like data, such as image recognition /segmentation and object detection in an image. RNNs uses loops to process a sequence of inputs with a state. RNNs are suitable for time-series analysis such as natural language processing. LSTM is a variant of RNN, particularly designed for tasks requiring context, such as language modeling, speech recognition, and sentiment analysis. The proposed implementation uses FNN with the training algorithm Levenberg-Marquardt. The architecture uses single layer that contains 100 neurons, activated by ReLU function, and regularization strength (lambda) set to 0.

6) APPLICATION OF SUBSPACE KNN

Random SubSpace is an ensemble learning method in which the features are randomly sampled with replacement for each weak learner with no limitation on the base learner. It means it can ensemble any learner. It is similar to Random Forest. The difference is that Random SubSpace only samples features and works with any learner whereas Random Forest samples both training set and features, and the base learner must be decision trees. SubSpace KNN is the Random SubSpace learning using the base learner as KNN. The proposed implementation uses 30 number of KNN learners as weak learners with the subspace dimension set to 512. Each KNN learner uses one nearest neighbor according to Euclid distance metric.

IV. EXPERIMENTAL RESULTS

In this section, the experimental results are provided for the feature extraction and classifier methods and the problem types. Two problems, namely letter recognition and person recognition, are studied using the suggested parameter values, which are given in the Section D of the Chapter III. No hyperparameter optimization is performed in the algorithms. However, the best parameter values are empirically found. The highest accuracy scores and their main parameters' values for the top 3 learners are summarized as follows: KNN classifier with K = 1, SubSpace KNN classifier with the subspace size 512 and the number of learners 30, ANN classifier with 1 layer having 100, 1.000, 5.000 and 10.000 neurons respectively. Although the performance increased slightly from 1.000 neurons to 5.000 neurons, the training time increased approximately 4 times. Similarly, the performance increased slightly from 5.000 neurons to 10.000 neurons, the training time doubled this time. Thus, it is decided to work

with 100 neurons. The methods are evaluated using accuracy scores with 10-Cross Validation where the accuracy scores achieved over 10 folds are averaged.

A. LETTER RECOGNITION

The accuracy scores achieved by the letter-classifiers according to 10-Cross Validation are shown in Table 2 and Table 3 for Fourier Transform and Wavelet Transform based feature extraction methods, respectively. According to the average results, KNN and SubSpace KNN perform similar, around 91%, for letter recognition using Fourier based feature extraction method. However, SubSpace KNN performs slightly better, achieving the best performance under the given parameter settings when compared to others. SVM and ANN also perform similarly, around 79%, where ANN is slightly better than SVM. As expected, the decision tree is the worst. For letter recognition rates, the letter 'Z' achieved the highest score with 95.4%, followed by the letter 'U' with 95.1% using SubSpace KNN.

In Wavelet domain, except the decision tree, the performances of all the classifiers tend to decline. The most dramatic decrease in the performance happens to be for KNN and ANN when compared to their variants in Fourier domain. However, SubSpace KNN achieves the best (89,5%), followed by the Random Forest (81,4%). Similarly, the lowest recognition rate comes from the Decision Tree algorithm.

Similarly, the letters 'U' and 'Z' are the most discriminative ones in Wavelet domain too. The results show that Fourier based classifications is better than the Wavelet variants. This is likely the result of freeform air-writing style where the subject does not obey any writing rule, such as font size, orientation, and handwriting style. Such freeform style generates more unmeaningful information in the time domain. Since Wavelet transformation preserves some of the information in both time and frequency domains, such details negatively affect the performance of wavelet-based classifiers. On the contrary, since Fourier transformation only preserves information in the frequency domain, it could better describe the letter variants in various orientations and sizes.

SubSpace KNN is expected to be better than KNN since Random SubSpace Learning is an ensemble of many KNNs as weak learners. Aggregating the decisions of many KNNs tend to overcome the overfitting problem as well as the high variance. This is similar to the relation between Decision Tree and Random Forest where the latter is an ensemble of many Decision Trees and, thus, performs better. Although ANN and SVM are strong learners for many problems, they have a similar weakness of gradient descent-based optimization of their parameters, e.g. weights. They may have better performance if the dataset belongs to a single person.

TABLE 2. Accuracy scores of fourier-based letter classifications according to cross validation.

Fourier Transformation						
Letter	DT	RF	KNN	S-KNN	SVM	ANN
A	37,8	82,7	91,7	90,8	78,9	84,5
B	68,2	75,2	93,4	93,4	74	83,4
C	54,1	85,9	90,1	93,6	85,3	87,5
Ç	44,7	87,1	92,6	91,3	78,7	82,6
D	33,3	80,1	89	91,3	78,6	80,6
E	37,5	73,1	88	89	74,1	79,3
F	35,3	76,3	88,1	90	72,3	72,3
G	45,5	84,7	92,3	92	80,8	81,4
Ğ	31,1	69,5	85,2	83,1	68,9	75,7
H	42,9	76,8	83,2	87,9	71,2	69,4
I	50	84,8	92	92,8	85,9	78,9
İ	40,4	87,9	90,7	92,4	82,1	78,3
J	55,6	87,1	92,8	92,8	81	83,7
K	50	82,2	91	88,9	73,2	77,1
L	44,4	90,2	91,9	91,9	87,1	80,7
M	51,6	83,5	89,4	91,2	77,6	81,2
N	24,4	81,7	89,9	89,3	77,5	80
O	55,9	78,1	92,3	92	77,4	82,5
Ö	25,7	84,9	87	86,4	76,5	77,1
P	34,2	87,8	91,7	93,2	79,7	80,5
R	33,3	84,5	90,2	92	77	81
S	40	80	90,8	92,5	81,9	77,5
Ş	34	86,8	91	90,4	79,4	77,2
T	20,7	86,7	90,8	93,2	75,1	81,8
U	40,9	82,9	93,5	95,1	78,5	80,6
Ü	24,2	84,7	83,9	88,3	74,2	74,2
V	39,4	85,4	92,8	94,1	83,5	80,7
Y	36,2	82,9	91,2	93,6	77,9	79,7
Z	35,3	88,1	95,4	94,9	86,5	83,6
GENERAL	**39,7**	**83,2**	**90,5**	**91,5**	**79**	**79,8**

B. PERSON RECOGNITION

The accuracy scores achieved by the subject-classifiers according to 10-Cross Validation are shown in Table 4 and Table 5 for Fourier Transform and Wavelet Transform based feature extraction methods respectively. According to the average results, SubSpace KNN performs the best, which has an around 93% accuracy ratio, for subject recognition using the Fourier-based feature extraction method when compared to others. Interestingly, KNN and ANN perform similarly, around 76.5%, where ANN is slightly better than KNN. As expected, the decision tree is the worst. The recognition rates of some subjects (#1, #3, #9, #12, #15) are higher than the average, indicating that these subjects have strong personalized writing styles associated with themselves. This helps the algorithm to discriminate them from the others.

Due to similar reasons, the Wavelet-based subject-classifiers tend to decrease in performance. Similarly, SubSpace KNN is the best of all due to the aforementioned

TABLE 3. Accuracy scores of Wavelet-based letter classifications according to cross validation.

Wavelet Transformation						
Letter	DT	RF	KNN	S-KNN	SVM	ANN
A	40,8	79,5	69,9	86,9	60,4	64
B	45,8	77,7	66,8	85,3	61,8	69
C	55,5	89,3	90,9	93,3	75,2	78,9
Ç	47,9	81,6	81,3	92,9	58,9	73,7
D	48,3	80,3	76,6	86,4	61	77,7
E	39,2	72,5	58,9	85,8	51,8	54
F	30,7	78,7	68,1	89,1	53,8	62,3
G	48,7	85,8	81,1	90	64,9	70,5
Ğ	34,5	69,2	49,5	84,9	57,2	62,2
H	37,9	77,9	63,2	83,5	53,2	63,5
I	47,3	86,1	91	92,5	79,9	81
İ	37,4	76,2	71,2	84,5	64,3	64
J	55,6	86,5	85,1	93,4	69,4	76,3
K	44,6	82,5	71,7	89,8	55,1	65,4
L	48,5	86,6	89,1	93,6	75,4	77
M	38,8	79,1	70	90	59,4	70
N	42,8	81,1	75,2	91,5	60,6	68,2
O	41,4	76,6	81	87,1	65,3	70,4
Ö	34,2	76,5	44,9	83,8	55,1	64,1
P	49,9	87,5	78,7	93,5	69,9	74,5
R	47,7	83	77,3	91,1	50,3	67,5
S	47,8	87,5	86,1	92,5	70,8	79,2
Ş	40	78,3	68,5	89,9	58,9	67,3
T	45,9	84,8	76,9	88	73,6	74,2
U	46,4	89,4	86,3	94,3	74,1	79
Ü	35,3	67,8	48,9	82,2	42,8	46,1
V	50,1	85,7	87,4	90,6	71,9	73,8
Y	46,7	84,5	79,2	91,2	67,5	74,9
Z	46,9	84,1	89,2	94,3	70,4	77,9
GENERAL	**44**	**81,4**	**75,1**	**89,5**	**63,6**	**70,1**

characteristics, equally applicable to subject recognition. It achieves 88.8% accuracy ratio.

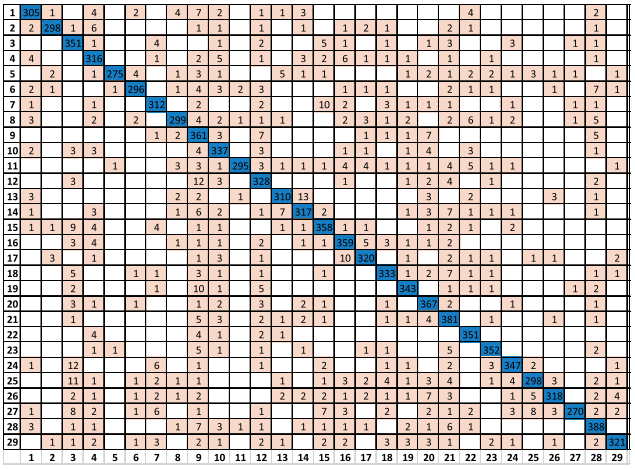


FIGURE 4. Confusion matrix for letter recognition using SubSpace KNN with fourier transform.

TABLE 4. Accuracy scores of fourier-based subject classifications according to cross validation.

Fourier Transformation						
Person	Decision Tree	Random Forest	KNN	S-KNN	SVM	ANN
1	63,7	91,6	96,6	98,1	94,4	89,4
2	41,3	66,6	86,8	88,1	79,7	78,9
3	48,8	79,8	93,7	94,2	88,1	83,5
4	32,1	57,5	85,9	88,3	77,4	74,8
5	16,7	49,6	77,8	90,2	73,6	67,9
6	31,9	62,0	89,3	95,5	85,5	72,1
7	29,9	60,3	91,6	93,7	81,7	77,6
8	31,5	63,1	92,3	94	82,4	78,4
9	38,8	79,6	94,2	95,5	91,3	86,8
10	20,3	48,9	75,4	87,5	76,7	65,7
11	44,0	79,3	92,7	93,7	87,3	82,9
12	42,7	80,7	93,5	97,1	89,1	83,8
13	60,9	87,1	89,7	94,6	91,5	84,5
14	32,5	58,3	68,6	82,1	76,1	70,2
15	71,9	95,6	99,3	99,4	98,3	92,9
GENERAL	**41,7**	**72**	**88,6**	**93,1**	**85,6**	**79,7**

C. DISCUSSION

To ensure a fair comparison with state-of-the-art methods, the most relevant studies on air-writing letter recognition have been highlighted. No studies on person recognition are included in the table, as there is a lack of literature on this topic. Thus, Table 6 presents only the accuracy scores of recent studies on air-writing letter recognition using IMU sensors (6 DoF or 9 DoF), ranging from 83.2% to 97.95%. However, these studies differ in dataset assumptions (e.g., multi-user, balanced, freeform) and performance evaluation methods (e.g., validation set, cross-validation). Dataset preparation significantly impacts performance. Fixedform styles, guided by templates, yield simpler, more recognizable samples, while freeform styles allow unrestricted writing, introducing higher variability. The imbalanced, multi-user, freeform dataset poses additional challenges.

While deep learning achieved the highest accuracy (97.95%) on a balanced, fixed-form dataset, and DTW with KNN reached 95.5% on a distinct test set, the proposed system demonstrates superior performance under cross-validation on an imbalanced, free-form, multi-user dataset.

Note that the proposed application here is suggested for special groups (kids, disabled people, etc.) to monitor their

TABLE 5. Accuracy scores of wavelet-based subject classifications according to cross validation.

Wavelet Transformation						
Person	Decision Tree	Random Forest	KNN	S-KNN	SVM	ANN
1	55,3	97,4	92,7	96,9	90,1	89,2
2	2,5	74,7	70,8	88,9	78,3	65,8
3	18,3	83,3	79,8	89,6	77,7	76
4	3,3	63,3	76,3	82	72,2	65,5
5	48,8	78,8	71,2	87,6	63,3	72,4
6	45,9	83	72,1	88,3	71,9	76,2
7	21,6	75	77,6	88,8	69,2	70,3
8	23,4	73,4	81,1	88,6	70,1	73,6
9	51	87,1	89,2	94	83,5	83,4
10	31,5	70,4	55	79,1	58	65,2
11	36	81,8	71,9	86,5	66,8	70,3
12	42,8	85,3	86,2	93,4	79,3	81,9
13	56,1	90,5	72,2	90	77,9	84,2
14	35,6	64	56,7	75,2	59,5	69
15	85,9	96,2	94,2	99,3	96,5	91,8
GENERAL	**40**	**81,3**	**76,4**	**88,8**	**74,2**	**76,6**

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	724	4		2	1		1	1	3	1			1		
2	4	422	8	15			3	2	8	3	6	1	1	1	5
3	2	6	490	6		3	1	3			3	1			5
4	12	11	6	476	1	2		6	4		1	5	7		8
5	1		1	6	582	13	2	5	1	3	5	3	10	8	5
6	1	2		1	2	653		3	1	2		4	12	3	
7		5	5	3		1	477	6	3		1	4	1		3
8	4	3	2	8	3		7	626	2	1	1	3	1	2	3
9	2	5	1	4			1	3	638	4	3	6	1		
10	8	3	1	1	6	13	2	8	3	687	2	12	22	17	
11	2	5	7	3	3		6	6	3	3	716	8		1	1
12			1	1	3	3	1	1	4	5	4	864	2	1	
13	6			1	3	27	2	1	3	11		2	1091	6	
14	14	2		2	7	19	4	6	5	22	2	8	26	535	
15		1		2	1										705

FIGURE 5. Confusion matrix for person recognition using SubSpace KNN with fourier transfor.

cognitive and motor development. Thus, this application considers air-writing as an experiment, conducted by a trainer, and poses some limitations, such as writing letters in a start-stop manner. Thus, the application in this form is sensitive to unplanned hand positions. Also, a limited device (6 DoF) was used, which is overshadowed by newer sensors currently. More advanced motion sensors having 9 DoF features may provide better results in future studies. Also, the classical machine learning algorithms may be replaced by deep

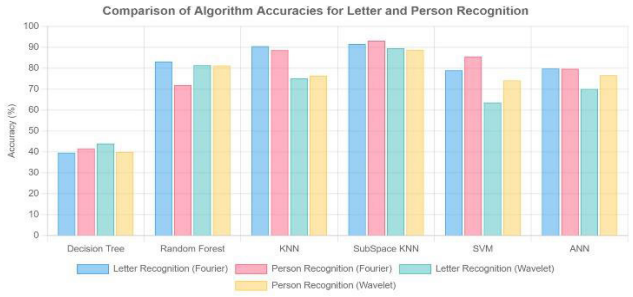


FIGURE 6. Comparison of models for letter recognition and person recognition using fourier and wavelet transform.

TABLE 6. Accuracies of letter based recognition: Comparison of some recent studies.

Study	IMU Type	Feature Extraction	Model	Dataset / Features	Validation/ Accuracy
[39] 2022	9DoF	Learnt	1D-CNN	custom / freeform balanced, multi-user	test set/ 91.3
[39] 2022	9DoF	2D-Trajectory Alignment	DTW+KNN	custom / freeform balanced, multi-user	test set/ 95.5
[40] 2021	9DoF	Template Matching	DTW	custom / freeform balanced, single-user	train set/ 84.6
[41] 2022	6DoF	GADF + Learnt	CNN (DenseNet)	public / fixedform balanced, multi-user	crossval/ 88.4
[50] 2020	6DoF	Learnt	1D-CNN	custom / fixedform balanced, multi-user	crossval/ 83.2
[42] 2024	6DoF	Learnt	1D-CNN-BiLSTM	public / fixedform balanced, multi-user	crossval/ 88.05
[4] 2018	6DoF	Fourier Coefficients	Random Forest	custom / fixedform imbalanced, single-user	test set/ 91.10
[43] 2022	6DoF	Learnt	CNN	custom / fixedform balanced, multi-user	test set/ 97.95
Proposed	6DoF	Fourier + Wavelet T.	SubSpace KNN	custom / freeform imbalanced, multi-user	crossval/ 91.50

learning algorithms in future studies for the text recognition problem, rather than letter recognition only.

V. CONCLUSION

This study jointly explores letter recognition and person recognition in air-writing to analyze the relationship between individuals and their freeform air-writing styles. To the best of the authors' knowledge, person recognition from freeform air-writing letters through IMU sensors is first introduced here. In this context, individuals freely write letters of any size in the air, and the application segments and identifies both the letters and the individuals. This combined analysis of air-writing data introduces new possibilities, suggesting air-writing as a tool for developing and assessing the cognitive and psychomotor abilities of individuals under special care, such as children, people with disabilities, and students.

For instance, a lack of consistency or significant variations in writing style may indicate delays or irregularities in fine motor skills or cognitive processes. Such observations can provide valuable insights into a child's developmental needs. This application contributes to these objectives by offering a practical framework for analyzing air-writing patterns and supporting the identification of cognitive and motor development indicators.

In this study, a dataset with multiple labels (letter and person) was also generated using an IMU-integrated wearable glove. The dataset contains data for 10.401 letters from 15 individuals. The dataset is unique from a few perspectives. It is imbalanced and contains freestyle data on Turkish alphabets. To the best of the authors' knowledge, there is no publicly available freeform air-writing dataset for the Turkish alphabet.

In this study, six machine learning algorithms (Decision Tree, Random Forest, KNN, S-KNN, SVM, and ANN) were compared for letter and person recognition using two feature extraction methods (Fourier Transform and Wavelet Transform). To the best of the authors' knowledge, Fourier and Wavelet features are first used here for freeform air-write recognition through IMU sensors. Furthermore, the comparison study of both letter and person recognition in air-writing contributes to the field. The SubSpace KNN algorithm was applied for the first time to both tasks and showed the highest accuracy rates. Fourier Transform outperformed Wavelet Transform in this context. The results indicated an over 93% association between individuals and their writing styles, with approximately 90% accuracy in letter recognition, independent of the writer. This research highlights the potential of air-writing for both letter and person recognition.

DATA AVAILABILITY

The dataset is publicly available on the GitHub repository: https://github.com/zyetgin/Air-Writing-Dataset

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